TEAM BRIEF

Smart Employee Onboarding & Identity Service

An automated system that takes a new hire from offer acceptance to a fully ready Day 1, built on AWS.

Evaluation date	4 November 2026
Team	Team Amit (F13 Technologies internship)
Platform	AWS (Lambda, DynamoDB, Cognito, Step Functions, S3, SES, SNS, API Gateway, CloudFront)
Repository	hrms-onboarding-service (GitHub)
Read time	About 20 minutes. Read it fully once, then pick your work package.

1. The project in plain words

When a company hires someone, a lot of small jobs must happen before Day 1: create their login, collect their documents, give them a laptop, make them accept company policies, and introduce them to their manager. Today this is done by hand through emails and spreadsheets, so things get forgotten and nobody can see who is stuck.

We are building a system that does all of it automatically and shows everyone the status. It is the **identity spine** of a bigger HR system: it creates the one official employee record that every other HR service will refer to later.

What the finished system must do

- Create an employee profile and a login the moment a new hire is registered.
- Run four stages in order: Document Collection, IT Provisioning, Policy Sign-off, Manager Intro.
- Send reminder emails automatically when a stage is not finished after 24 hours.
- Let the new hire upload documents safely, and check them automatically.
- Notify HR when all documents are received and verified.
- Show progress in a clean web app: one view for the new hire, one for the HR admin.
- Be cheap to run: our cost estimate is for 50 onboardings per month.

2. The new hire's journey, step by step

Step	What happens	Who or what does it
1	HR registers the new hire using a form (name, email, department, role, manager, joining date, employment type).	Web form, then API Gateway
2	A unique employee_id (UUID) is generated and the profile is saved.	Lambda, DynamoDB
3	A login account is created with a temporary password and an invite email with a login link is sent.	Cognito, SES
4	The onboarding workflow starts for this employee.	Step Functions
5	Stage 1, Document Collection: the hire uploads ID proof, degree certificate and signed offer letter. Files are checked and stored encrypted.	S3, Lambda
6	When all 3 documents are verified, HR gets a notification.	SNS
7	Stage 2, IT Provisioning: a laptop and accounts are assigned (simulated by default, real integration optional).	Lambda, DynamoDB
8	Stage 3, Policy Sign-off: the hire reads and accepts the company policies.	Web app, API
9	Stage 4, Manager Intro: the manager and hire both get an introduction email.	SES
10	Status becomes Day 1 Ready. The hire logs in on Day 1 and sees everything complete.	Web app
Any time	If a stage is not done after 24 hours, a reminder email is sent and the workflow keeps waiting.	Step Functions, SES

3. Architecture and why each service is used

Service	Its job in our project	Why this one
API Gateway	The front door: every web request goes through it.	Managed, secures routes with Cognito login.
Lambda	Small functions that do each job (create profile, validate file, send email).	Pay only when it runs, nothing to maintain.
DynamoDB	Stores employees, stage status, documents, equipment.	Fast, serverless, cheap at small scale.
Cognito	Creates and manages logins; separates HR admins from new hires using groups.	Built-in secure authentication.
Step Functions	The workflow engine. Knows which stage comes next and waits for humans.	Visual execution history, built-in timeouts and retries.
S3	Stores uploaded documents (encrypted) and hosts the website files.	Durable and cheap.
CloudFront	Serves the website over HTTPS in front of S3.	Cognito login needs HTTPS; also faster.
SES	Sends all emails (invites, reminders, introductions).	Low-cost email.
SNS	Notifies HR when documents are complete.	Simple publish and subscribe.
IAM	Controls who and what can touch each resource.	Least privilege keeps us safe.
CloudWatch	Logs and alarms.	First place to look when something breaks.

The callback pattern (the most important idea)

A normal program cannot wait three days for a person to upload a file. Step Functions can. At each stage, the workflow starts a Lambda, hands it a **task token** (like a numbered ticket), and pauses. When the person finishes the task, our API sends that token back and the workflow continues. If 24 hours pass with no token, the stage times out, a reminder is sent, and the workflow goes back to waiting.

Simulated and real modes

External systems (IT ticketing, chat) sit behind small adapter functions with a setting **MODE = simulated or real**. Simulated mode uses our own DynamoDB tables so the demo can never break. Real mode calls a free real service (for example GitHub Issues as the IT helpdesk and a Slack webhook for the manager introduction). The workflow does not change either way.

4. Data model (DynamoDB)

Table	Keys	Main attributes
Employees	PK: employee_id (UUID)	full_name, email, department, role, manager_id, joining_date, employment_type, status, execution_arn, created_at
OnboardingStages	PK: employee_id, SK: stage	status (pending, in_progress, complete), started_at, completed_at, reminder_count, task_token
Documents	PK: employee_id, SK: doc_type	s3_key, status (missing, uploaded, verified, rejected), file_size, content_type, validated_at
Equipment	PK: asset_id	type, model, status (available, assigned), assigned_to

A secondary index on Employees by **status** lets the HR dashboard list the pipeline quickly. Stage names: document_collection, it_provisioning, policy_signoff, manager_intro. Document types: id_proof, degree_certificate, signed_offer_letter.

5. API contract (agreed interface)

Everyone builds against this table. If you need a change, ask the lead first so nobody's work breaks.

Method and route	Purpose	Who can call
POST /employees	Register a new hire and start onboarding	HR admin
GET /employees	List all hires with stage status (HR dashboard)	HR admin
GET /employees/{employee_id}	Get one employee profile	HR admin, that employee
GET /employees/{employee_id}/progress	Progress of all four stages	HR admin, that employee
POST /employees/{employee_id}/documents/upload-url	Get a secure one-time upload link for a document type	That employee
GET /employees/{employee_id}/documents	Status of each required document	HR admin, that employee
POST /employees/{employee_id}/policies/sign	Record policy acceptance with a timestamp	That employee
POST /employees/{employee_id}/stages/{stage}/complete	Manually mark a stage done (for HR override and testing)	HR admin

Agreed defaults: allowed upload types are PDF, JPG and PNG; maximum size 5 MB; region us-east-1; resource names follow **hrms-dev-<component>** (for example hrms-dev-documents-bucket); all documents encrypted at rest.

6. Work packages: choose what you want to build

These are the pieces of the project. Nothing is pre-assigned. Read them all, tell the team which one or two you want, and we will confirm so nothing overlaps. Every package has a clear input, a clear output and a 'done when' checklist, so you can build and test it on its own.

WP1. Employee record and identity	
What	Lambda that creates the profile with a UUID, DynamoDB tables, Cognito user pool with HR and Employee groups, invite email via SES.
Input and output	Form data in; employee row, Cognito user and invite email out.
Done when	Profile saved with every field; hire receives an invite; duplicate emails rejected.
Skills	Lambda, DynamoDB, Cognito, SES

WP2. Workflow engine	
What	The Step Functions state machine with four stages, callback waits, 24-hour timeout and reminder loop, plus stage status tracking in DynamoDB.
Input and output	employee_id in; stage updates and completed workflow out.
Done when	Four stages run in order; timeout triggers a reminder; execution history shows each run.
Skills	Step Functions, Lambda, DynamoDB

WP3. Document collection	
What	Presigned upload URLs, encrypted S3 bucket, validation of file type and size, per-document status, HR notification when all three are verified.
Input and output	employee_id and task token in; Documents rows, SNS email and workflow resume out.
Done when	Bad type or oversize file rejected; HR gets exactly one email when all three are verified.
Skills	S3, Lambda, SNS

WP4. Stage handlers	
What	IT Provisioning (equipment assignment), Policy Sign-off and Manager Intro Lambdas, with simulated and real mode switch.
Input and output	employee_id and task token in; stage marked complete out.
Done when	Both modes work; switching mode needs no workflow change.
Skills	Lambda, DynamoDB, SES, SSM

WP5. Progress and HR API	
What	API Gateway routes from the contract, Cognito authorizer, role-based access, progress endpoint.
Input and output	HTTP requests in; JSON responses out.
Done when	Every route in section 5 works; an employee cannot see another employee's data.
Skills	API Gateway, Lambda, Cognito

WP6. New hire portal (frontend)	
What	Premium, clean web app for the hire: login, progress bar per stage, document upload, policy sign-off.
Input and output	Uses the API contract.
Done when	Works on desktop and mobile; matches the shared design system.
Skills	HTML, CSS, JavaScript, S3, CloudFront

WP7. HR admin dashboard (frontend)	
What	Pipeline view for every hire, colour-coded status (pending, in-progress, complete), register-new-hire form.
Input and output	Uses the API contract.
Done when	Status colours are correct; live data from the API.
Skills	HTML, CSS, JavaScript

WP8. Documentation and demo	
What	Architecture and ER diagrams, Step Functions execution screenshots, cost estimate for 50 onboardings, demo video script and recording.
Input and output	Working system in; polished deliverables out.
Done when	All four required deliverables complete and consistent.
Skills	Diagrams, writing, screen recording

Your next step: post in Team Amit which one or two packages you would like to own and why. If anything in this brief is unclear, ask before you start building.

7. Access and security (IAM)

Real companies never give everyone full access. We practice that. Once you pick a package, you get an IAM role that lets you touch only your own resources.

Role	Can do	Cannot do
Platform admin (lead)	Everything, with MFA required	Nothing blocked
Module developer	Read and change only resources named for their package (for example hrms-dev-documents-*)	Change IAM, billing, or other people's resources
Frontend developer	Upload to the frontend bucket, manage CloudFront, read API and Cognito settings	Change backend resources
Read-only reviewer	View everything for review and demo	Change anything

Security rules for everyone

- Never put passwords, keys or tokens in GitHub. Secrets live in SSM Parameter Store.
- Turn on MFA for your account. Never use the root user for daily work.
- Give every Lambda only the permissions it needs, nothing more.
- Set a billing alarm at $5 on any account you create.

8. How we work together (GitHub)

Topic	Rule
Branches	main is protected. Work on feature/<package>-<short-name>, for example feature/wp3-upload-validation.
Pull requests	No direct pushes to main. Every change goes through a pull request with at least one review from the lead.
Commits	Short, clear messages that start with a type: feat, fix, docs, chore.
Folders	Each package has its own folder: services/, frontend/, infra/, docs/. Stay inside yours.
Issues	Each work package is a GitHub Issue. Update it as you go.
Communication	Daily short update in the Team Amit space: done, doing, blocked.
Changes to the contract	Ask the lead first. The contract lives in docs/api-contract.md.

9. Deliverables and timeline

Deliverable	Notes
Working onboarding flow	Registration to Day 1 login, with Step Functions execution history screenshots.
ER diagram	DynamoDB employee schema (section 4 is the base).
Video demo	Full new-hire journey from form fill to Day 1 login.
Cost estimate	For 50 onboarding events per month, with assumptions listed.
Professional frontend	Clean, light, premium look with progress bars and colour-coded status.
Documentation	README, architecture notes and this brief kept up to date.

Date	Milestone
26 to 27 Oct	Read this brief, pick work packages, set up AWS account and GitHub access
28 Oct	Contract frozen (tables, routes, events); everyone starts building
29 Oct to 1 Nov	Build and test your own package
2 Nov	Integration: all packages connected and tested together
3 Nov	Polish, screenshots, diagrams, cost estimate, demo recording
4 Nov	Evaluation

Definition of done for any package

- It works against the contract and passes its own 'done when' checklist.
- No secrets in the code. Resources follow the naming rule.
- It has a short README explaining what it does and how to test it.
- It was reviewed and merged through a pull request.

10. Glossary

Term	Meaning
Lambda	A small piece of code that runs only when triggered.
DynamoDB	AWS's fast key-value database.
Cognito	AWS's login and user management service.
Step Functions	A visual workflow service that runs steps in order and can wait.
Task token	A ticket the workflow hands out and waits for before continuing.
Presigned URL	A temporary link that lets a user upload one file straight to S3.
IAM role	A set of permissions attached to a person or a service.
Adapter	A small wrapper that lets us swap a simulated system for a real one.